Intrusion Report – Feb 21, 2026

Team: Team 32

Environment: ISEAGE

Report Timeframe: Feb 20, 2026 15:09 – Feb 21, 2026 15:29

Prepared By: White Team

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1. Executive Summary

During the reporting period, we detected **malicious activity consistent with Red Team operations**, including SSH brute-force attempts, brief authenticated access, and **unauthorized file manipulation in the `/root` directory**.

Because these actions were **detected immediately through IDS monitoring**, our team was able to **identify and remove attacker artifacts prior to official notification**. We confirmed with the **Physical White Team** that this action was **authorized**, provided detection occurred first.

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2. IDS Activity Report

We continued monitoring all systems using **OSSEC integrated with ELK**, which provided real-time alerts for:

- File integrity changes
- Authentication failures
- Privileged directory access

During this window:

- Over **1,200 alerts** were generated
- Rule Levels **5–7** were triggered repeatedly
- Alerts correlated strongly with SSH activity and root directory changes

The IDS allowed us to reconstruct attacker actions with high confidence.

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3. SSH Brute Force & Authentication Failures

Observed Behavior

- Sustained SSH brute-force attempts from:
  - `49.10.219.164`
  - `12.110.206.226`
  - `12.110.206.140`
- Login attempts against:
  - `root`
  - `cdc`
  - non-existent users
- Repeated `sshd: authentication failed` events

Evidence

Feb 21, 2026 @ 13:44:30.983	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:44:26.982	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:44:24.983	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:39:20.890	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:39:18.903	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:39:14.885	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:32:06.819	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:32:04.819	www	cdc	49.10.219.164	sshd: authentication failed.
Feb 21, 2026 @ 13:32:00.818	www	cdc	49.10.219.164	sshd: authentication failed.

Feb 21, 2026 @ 14:18:11.005	mgmt	cdc	12.110.206.140
Feb 21, 2026 @ 14:09:12.803	mgmt	cdc	12.110.206.226
Feb 21, 2026 @ 14:00:04.625	mgmt	cdc	49.10.219.164
Feb 21, 2026 @ 14:00:04.603	mgmt	root	49.10.219.164
Feb 21, 2026 @ 13:55:40.487	mgmt	cdc	12.110.206.226
Feb 21, 2026 @ 13:52:34.404	mgmt	cdc	12.110.206.226
Feb 21, 2026 @ 13:43:39.642	mgmt	cdc	49.10.219.164
Feb 21, 2026 @ 13:43:39.635	mgmt	root	49.10.219.164
Feb 21, 2026 @ 13:38:31.400	mgmt	cdc	199.100.16.101

Most attempts failed and were **automatically blocked** by firewall-drop Active Response rules.

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4. Unauthorized Root-Level File Modifications

Critical Detection

IDS detected **multiple unauthorized file events under** `/root` on the management host:

- `/root/.ssh/authorized_keys`
  - **Added**
  - **Modified (checksum change)**
  - **Deleted**
- `/root/krb.tgz`
  - Added and later deleted

This sequence strongly indicates **post-authentication activity and attempted persistence via SSH key injection**.

Evidence

The rapid add → modify → delete cycle suggests the attacker was actively testing or cleaning up access.

Feb 21, 2026 @ 15:29:41.6…	mgmt	/root/.ssh/authorized_keys	deleted	File deleted.	7	553
Feb 21, 2026 @ 15:27:23.3…	mgmt	/root/krb.tgz	deleted	File deleted.	7	553
Feb 21, 2026 @ 15:24:47.9…	mgmt	/root/krb.tgz	added	File added to the system.	5	554
Feb 21, 2026 @ 15:24:47.9…	mgmt	/root/.ssh/authorized_keys	modified	Integrity checksum changed.	7	550
Feb 21, 2026 @ 15:24:47.9…	mgmt	/root/.ssh/authorized_keys	added	File added to the system.	5	554

5. Red Team Flag Activity

Detected Flags

The following Red Team flags were identified by the IDS:

- `/root/team32_mgmt-root.flag`
- `/root/team32_www-root.flag`

These files were:

- Added
- Detected by file integrity monitoring
- **Deleted shortly afterward**

Evidence

This behavior is consistent with **Red Team objective completion** followed by artifact removal.

Feb 21, 2026 @ 15:06:01.9...	www	/root/team32_www-root.flag	added	File added to the system.	5	554
Feb 21, 2026 @ 14:52:45.4...	mgmt	/root/team32_mgmt-root.flag	added	File added to the system.	5	554

1,239 hits

Feb 20, 2026 @ 15:09:35.538 - Feb 21, 2026 @ 15:09:35.538

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timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Feb 21, 2026 @ 15:08:54.1...	www	/root/test.txt	added	File added to the system.	5	554
Feb 21, 2026 @ 15:06:01.9...	www	/root/team32_www-root.flag	added	File added to the system.	5	554

6. Coordination With Physical White Team

After detecting the unauthorized SSH key modification and Red Team flag placement, we **contacted the Physical White Team** for guidance.

Physical White Team confirmed that:

- If attacker artifacts (keys, flags, files) are **detected before official notification**
- And are **clearly identified via IDS**
- They **may be removed immediately**

Based on this guidance, the following were **intentionally removed**:

- Unauthorized `authorized_keys` entries
- Red Team flag files in `/root`

All removals were logged by the IDS and are included in this report for transparency.

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7. Active Response & Mitigation

Automated Actions

OSSEC Active Response triggered **firewall-drop rules**, blocking attacking IPs after repeated violations.

Effectiveness

- SSH brute-force activity was reduced
- Attacker persistence was disrupted
- No long-term backdoors were observed

Evidence

Feb 21, 2026 @ 14:09:16.590	Host Blocked by firewall-drop Active Response	sshd: Attempt to login using a non-existent user	12.110.206.226
Feb 21, 2026 @ 14:09:14.589	Host Blocked by firewall-drop Active Response	sshd: Attempt to login using a non-existent user	12.110.206.226
Feb 21, 2026 @ 14:00:36.240	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:44:43.066	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:44:12.379	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:39:34.986	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:38:55.846	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:35:21.546	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:32:22.813	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164
Feb 21, 2026 @ 13:31:50.290	Host Blocked by firewall-drop Active Response	sshd: brute force trying to get access to the system. Authentication failed.	49.10.219.164